

DESIGN FOR TESTABILITY (DFT)

Scan Path Insertion & ATPG Based Digital Circuit Testing

Comprehensive 7-Page Project Report

Project Area	VLSI / Digital IC Testing
Main Techniques	Scan Chain, Fault Modeling, ATPG
Target Circuit	Small RTL / Combinational Logic Circuit
Tools	Verilog HDL, Icarus Verilog, GTKWave
Academic Department	Electronics & Telecommunication (ENTC)

Abstract

Design for Testability (DFT) is a collection of techniques used to make digital integrated circuits easier to test after fabrication. This project studies scan-based DFT, in which ordinary flip-flops are converted into scan flip-flops and connected as a scan chain. Test patterns generated using Automatic Test Pattern Generation (ATPG) are applied to detect modeled faults. The project demonstrates scan insertion, shift/capture operation, fault detection and simulation-based verification.

1. INTRODUCTION

Modern digital systems contain a very large number of gates, registers and interconnections. Manufacturing defects can cause a circuit to behave incorrectly even when the RTL design is logically correct. Testing is therefore an important stage of VLSI design. DFT modifies the design so that internal circuit nodes become more controllable and observable.

In a conventional sequential circuit, an internal register may be difficult to control from primary inputs and difficult to observe at primary outputs. Scan testing solves this problem by providing a serial path through the storage elements. During test mode, flip-flops operate like a shift register. Test data can be shifted in, functional operation can be performed for capture, and the response can then be shifted out for analysis.

The project combines three major concepts: **scan path insertion**, **fault modeling** and **ATPG**. The complete flow can be demonstrated using free HDL simulation tools.

2. PROBLEM STATEMENT

Develop and verify a small digital circuit with DFT capability by converting its sequential storage elements into scan elements, connecting them into a scan chain, applying test patterns, and demonstrating detection of stuck-at faults through simulation.

3. OBJECTIVES

- Understand the purpose and importance of DFT in VLSI.
- Design a small RTL circuit suitable for scan-based testing.
- Understand scan-in, scan-out, scan-enable and clock signals.
- Create and verify scan-chain operation using Verilog.
- Model common stuck-at-0 and stuck-at-1 faults.
- Generate or apply ATPG-style test patterns.
- Compare functional-mode and test-mode behavior.
- Observe waveforms and document fault-detection results.

4. DFT ARCHITECTURE

A basic scan-based architecture contains the functional combinational logic, scan flip-flops, a scan-enable control, scan input and scan output. When Scan Enable is inactive, the circuit behaves normally. When Scan Enable is active, each scan flip-flop selects Scan In instead of its functional D input.

The simplified data selection principle is:

$$Q(\text{next}) = \text{Scan_Enable} ? \text{Scan_In} : \text{Functional_D}$$

5. SCAN CHAIN OPERATION

Shift phase: Scan Enable is set to 1. Test bits are applied serially at Scan In and move through the scan flip-flops on clock edges.

Capture phase: Scan Enable is set to 0. The functional circuit is allowed to respond to the applied test condition, and the resulting values are captured by the scan flip-flops.

Shift-out phase: Scan Enable is returned to 1. Captured responses are shifted through Scan Out and compared with the expected response.

6. TEST SIGNALS

Signal	Purpose
CLK	Clock used for shifting and capture
RESET	Initializes the sequential elements
SE / Scan Enable	Selects scan or functional operation
SI / Scan In	Serial test-data input
SO / Scan Out	Serial test-response output
Functional Inputs	Normal circuit inputs
Functional Outputs	Normal circuit outputs

7. BASIC BLOCK-LEVEL FLOW

RTL Circuit → Scan Insertion → Scan Chain → Fault Modeling → Test Pattern / ATPG → Simulation → Fault Coverage Analysis

8. IMPLEMENTATION METHODOLOGY

The project can be implemented in a free simulation environment using Verilog HDL. Icarus Verilog can compile the design and testbench, while GTKWave can display simulation waveforms.

Step 1 – Design the functional circuit

Write a small RTL circuit containing combinational logic and sequential elements.

Step 2 – Identify scan elements

Select flip-flops/registers that will participate in the scan path.

Step 3 – Add scan multiplexing

Use a 2:1 selection structure so each flip-flop receives either functional data or scan data.

Step 4 – Connect the chain

Connect the Q output of one scan element to the scan input of the next.

Step 5 – Add test controls

Provide Scan Enable, Scan In and Scan Out signals.

Step 6 – Build the testbench

Generate reset, clock, functional inputs and scan patterns.

Step 7 – Model faults

Introduce stuck-at faults at selected logic nodes for verification.

Step 8 – Run simulation

Compile the Verilog sources and inspect the resulting waveforms.

Step 9 – Analyze coverage

Record which modeled faults are detected by the applied patterns.

9. SOFTWARE / TOOLS

Recommended free tools are **Verilog HDL** for hardware description, **Icarus Verilog** for compilation and simulation, and **GTKWave** for waveform visualization. A text editor or VS Code can be used to create the source files.

10. PROJECT FILE STRUCTURE

A simple repository can contain: **rtl/** for design files, **tb/** for testbenches, **faults/** for fault models, **results/** for waveform screenshots and reports, and a **README.md** explaining how to run the simulation.

11. FAULT MODELING AND ATPG

A fault model is an abstraction used to represent possible manufacturing defects. The most common introductory model is the single stuck-at fault model. A node may be permanently stuck at logic 0 (SA0) or logic 1 (SA1).

For a fault to be detected, a test pattern must first **activate** the fault and then **propagate** its effect to an observable output or scan response. ATPG is the process of finding such input patterns automatically.

12. STUCK-AT FAULT EXAMPLE

Suppose a logic node normally can become either 0 or 1. To detect an SA0 fault, a test must attempt to make the good circuit node equal to 1. If the faulty node remains 0 and this difference propagates to an output, the fault is detected. Similarly, an SA1 fault requires an attempted value of 0.

13. SCAN-BASED TEST SEQUENCE

Phase	Scan Enable	Main Action
Reset	X	Initialize scan registers
Shift In	1	Load serial test pattern
Capture	0	Capture functional response
Shift Out	1	Unload captured response
Compare	1	Compare observed and expected response

14. EXPECTED SIMULATION OBSERVATIONS

During shift operation, Scan In should appear at Scan Out after the appropriate number of clock cycles, demonstrating correct scan-chain connectivity. During capture, the scan registers should store the circuit response. A modeled fault should cause a mismatch between the expected good response and the observed faulty response for at least one detecting pattern.

15. VERIFICATION AND RESULTS

Verification should be performed in two modes: functional mode and scan-test mode. Functional mode confirms that the original circuit still operates correctly. Scan mode confirms that test data can be loaded and unloaded through the chain.

16. SAMPLE RESULT TABLE

Test	Mode	Expected Observation	Status
Reset	Functional/Test	All scan registers initialize	PASS
Scan shift	Test	Serial pattern propagates through chain	PASS
Functional operation	Functional	Circuit produces expected output	PASS
Capture	Test	Response is stored in scan registers	PASS
Shift-out	Test	Captured response appears at Scan Out	PASS
SA0 test	Fault test	Detecting pattern creates response mismatch	PASS*
SA1 test	Fault test	Detecting pattern creates response mismatch	PASS*

*PASS indicates the intended simulation check; actual results should be updated with the user's measured waveform/fault-coverage data.

17. WAVEFORM CHECKPOINTS

- Clock edges must be continuous and correctly timed.
- Scan Enable must be high during shifting and low during capture.
- Scan In data should propagate one stage per active clock edge.
- Scan Out should show the shifted-out response in the expected order.
- Reset should place the scan registers in the defined initial state.
- Fault simulation should produce a distinguishable response when a fault is detectable.

18. ADVANTAGES OF SCAN DFT

Scan DFT improves controllability and observability of internal registers, simplifies sequential test generation, supports automated test equipment, and provides a structured approach to fault detection. It also makes debugging and manufacturing-test development more systematic.

19. LIMITATIONS AND FUTURE SCOPE

Scan insertion adds hardware such as multiplexing logic and scan routing. Test operation also introduces additional shift cycles and may increase test time. Large industrial designs require more advanced DFT features such as multiple scan chains, compression, boundary scan, memory BIST, logic BIST and sophisticated ATPG algorithms.

Future work can extend this academic project by increasing circuit size, implementing multiple scan chains, measuring fault coverage automatically, adding a fault simulator, generating ATPG patterns programmatically, and comparing different scan architectures.

20. CONCLUSION

This project demonstrates the fundamental idea of Design for Testability using scan-chain architecture and ATPG-oriented fault testing. By adding scan access to sequential elements, internal states become easier to control and observe. The simulation flow provides a practical understanding of shift, capture and shift-out operations and shows how stuck-at faults can be detected using suitable test patterns. The project forms a useful foundation for advanced VLSI testing and DFT concepts.

21. VIVA / PRESENTATION POINTS

- **What is DFT?** Design techniques that make a circuit easier to test.
- **Why scan chain?** To improve controllability and observability of internal storage elements.
- **What is ATPG?** Automatic generation of test patterns intended to detect modeled faults.
- **What is a stuck-at fault?** A model in which a node behaves as permanently 0 or permanently 1.
- **What are scan-in and scan-out?** Serial ports used to load test data and unload test responses.
- **What is scan enable?** A control signal that selects scan shifting instead of normal functional data.
- **What is fault coverage?** The proportion of modeled faults detected by the applied test patterns.

22. REFERENCES

1. M. L. Bushnell and V. D. Agrawal, *Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits*.
2. N. Jha and S. Gupta, *Testing of Digital Systems*.
3. IEEE terminology and introductory literature on Design for Testability and Automatic Test Pattern Generation.
4. Verilog HDL simulation documentation and Icarus Verilog / GTKWave user documentation.

23. Waveform Result:



